

Full Session Summary — ED Workflow Coordination

End-to-End Session Summary — ED Workflow Coordination (2025-08-16 14:42 UTC)

1) Context & Goal

- Shift the project from monolithic clinical prediction to a **workflow coordination system** that reduces cognitive load (paging, reminders, order pre-population), **without** making diagnoses.
- Architecture mandate: **small, composable skills** with a coordination layer so the system can **co-evolve** with hospital needs; avoid overfitting and black-box drift.

2) Timeline of Key Events (condensed)

- **Initial failure:** Your ``ed-ops-pipeline-v2b.ipynb`` had the entire notebook JSON pasted in a `*single code cell*` → nothing could run.
- **Fix:** Reconstructed as ``ed-ops-pipeline-v2b.reconstructed.ipynb`` with real cells.
- **Papermill error:** `*No kernel name found in notebook*`.
- **Fix:** regenerated notebook with valid kernel metadata.
- **Dataset QA push:** Hardened **ID overlap** detection and class **prevalence** checks; recognized site uses **Fall-ID** and requested ``patient_id`` mapping; confirmed synthetic-only runs for now.
- **Architecture pivot:** From CatBoost-like single model to **skills network** + **governor** + **PatientStateManager** (per-encounter states).
- **Kaggle path issues** (``/mnt/data`` writes failing) and **CONFIG missing** → ``NameError: CONFIG``.
- **Fix:** path guard to ``/kaggle/working``; bootstrap ``CONFIG`` if missing.
- **Equipment tracking (QR) resurfaced:** Found your QR pack and mapping; re-integrated **ResourceTracker** with reservations/TTL, cleaning/maintenance, battery/docking.
- **State bug:** per-event replay recreated ``WorkflowState()`` each loop → lost context.
- **Fix:** **PatientStateManager** keyed by ``encounter_id``.
- **ICU constraints realism:** STEMI fast track is solved; **sepsis holds** clog the ED. Adjusted priority logic and added lingering safety prompts.
- **Assumption corrected:** "Night shift CT is slower" → **not true** at your site. We fixed CT follow-up to **60 min** (shift-neutral).
- **TinyCritics attribute** & other NameErrors: removed stale references; centralized thresholds in ``CONFIG``.
- **UI file missing:** ``EDEquipmentTrackerPro.tsx`` write failed due to missing dir.
- **Fix:** `mkdir` guards + path resolver.
- **Acceptance/alert metrics:** early run ``accepted=0/3``; later ``accepted=9/20`` → **acceptance ~0.45**, **fatigue ~0.55**; baseline recorded in ``metrics.json`/`emissions.json``.

- **Language barriers**: online translation is a privacy risk → kept **offline phrase-card hooks** only.
- **Ethics**: **age/sex** removed from decision logic; scores remain display-only.
- **Release cadence**: ``ED_Pipeline_v4_KaggleFix.ipynb`` → **patched** to ``ED_Pipeline_v4.1.ipynb`` with an **audit cell** to prevent regression.

3) Bugs & Failures (with fixes)

- **Notebook JSON in a cell** → reconstructed notebook with valid cells.
- **Kernel metadata missing** (papermill) → injected proper kernel name.
- **`CONFIG` undefined** → bootstrap guard + defaults.
- **FileNotFoundError** for UI artifact (TSX) → directory creation before write.
- **`TinyCritics.benefit\_bias` AttributeError** → removed hard-coded class attr, used ``CONFIG``.
- **Duplicate nagging / alert fatigue** → **anti-spam gate** (cooldown, repeat cap, snooze-after-accept).
- **Incorrect CT night-delay rule** → removed; fixed **CT follow-up at 60 min**.
- **State loss in replay** → **PatientStateManager** per encounter.

4) Architecture — “Smaller models that can grow”

Before: single learner on synthetic sequences; temporal noise, brittle features. **Now:** **skills network** + **coordination layer**:

- **WorkflowState** + **PatientStateManager** (chronological, per-encounter).
- **Skills** (tiny units): CT/30m follow-ups, serial troponin + second ECG, troponin delta **display-only**, snacks/hydration prompts, consult follow-ups, SOP surfacing, lingering re-checks.
- **ResourceTracker core**: QR scans update location/status; reservations with TTL; cleaning/maintenance; docking/charge; **analytics** hooks.
- **Safety & governance**: suggestions-only; anti-spam gate; **age/sex not used** in decision rules; “shadow” learning mode only.
- **Co-evolution**: new hospital needs → add/retire skills; measure **acceptance**/**fatigue** before promotion.

5) Domain Expertise — loss and reintegration

Lost along the way: QR logistics, SOP access, sepsis-hold reality, language barriers, display-only risk awareness (MEWS/qSOFA/HEART/etc.), echo after conversion preference, pregnancy cautions, allergy cross-reactivity levels. **Reintegrated as code & policy:**

- **Equipment**: QR, maintenance, cleaning, battery/docking, conflict/TTL.
- **SOPs**: registry + search + **usage logging**.
- **Sepsis-hold ops**: lingering prompt (>2h), snacks/hydration (≥4h, non-NPO).

- **Chest pain**: serial troponin, **second ECG with second troponin**, **delta review (display-only)**; echo preference after (electro)conversion.
- **Allergy logic**: penicillin↔cephalosporin warning; contrast premed; NSAID alternatives.
- **Pregnancy**: imaging warnings; not a decision driver for emergencies.
- **Language**: offline **phrase-card** hooks only.
- **Risk scores (display-only)**: **MEWS**, qSOFA, SOFA■lite, HEART, Marburg, GRACE■proxy with **age/sex excluded** from decisions.

6) Data QA & Leakage

- Hardened **ID detection** (MRN, encounter, account, chart, episode, case, *_id, *_key, etc.).
- **Overlap table** logic to ensure **no patient/encounter leakage** between train/val/test.
- **Prevalence table** to compare class mix across splits.
- Real vs synthetic: current runs remain **simulation**; wiring ready for real ED logs when available.

7) Features now in the core

- **ResourceTracker** with analytics and finder (`find_assets`) + status board JSON.
- **SOP surfacing** + **SOP_USAGE** logging.
- **Troponin** serial + delta prompt (display-only) + **second ECG** reminder.
- **Follow-ups**: **CT > 60m**, **Cath/Endo > 30m**.
- **Lingering** and **snacks/hydration** skills.
- **Anti■spam** gating to avoid fatigue.
- **Audit cell**: `domain_requirements_audit()` → PASS/FAIL table.

8) Checklists & Governance

- **ED Research Checklist** and **Clinical Requirements spec** are canonical.
- **Regular audit**: each run of `ED_Pipeline_v4.1.ipynb` executes `domain_requirements_audit()` to detect regressions (equipment, SOP, troponin, lingering, scores, anti-spam, ethics).

9) Measured outputs seen

- Early run: suggestions=3, accepted=0 → **acceptance 0.0**, **fatigue proxy 1.0**.
- Later run: suggestions=20, accepted=9, rejected=11, **clicks_saved=9**, **seconds_saved=90**, **acceptance 0.45**, **fatigue 0.55**.
- CSV scan: “No train/val/test under /kaggle/input” → synthetic-only run confirmation.

10) Current notebook set

- **ed-ops-pipeline-v2b.reconstructed.ipynb** — recovered from the JSON-in-cell issue.
- **ED_Pipeline_v4_KaggleFix.ipynb** — base with Kaggle-safe paths.
- **ED_Pipeline_v4.1.ipynb** — domain patches + audit.
- Helper: **patch_to_v4_1.py** — builds v4.1 from v4 in Kaggle.

11) Remaining gaps (explicit)

- Multi-patient **status board UI**; SOP checklist/flow renderer with step tracking.
- Equipment **usage analytics** dashboards; ROI/time-to-find reports.
- **Data validation**: out-of-order/duplicate events; clinical range checks; required fields.
- **Integration tests**: multi-patient contention; emergency override pathways; maintenance notifications to radiology/biomed.
- **Tuning**: acceptance ≥ 0.65 ; fatigue ≤ 0.30 ; ≤ 1 repeat/encounter.

12) Next Steps (actionable)

1. Add **EXPEDITE_LABS** and **FOLLOW_UP_CONSULT** skills with tests. 2. Implement **status board UI** (overdue assessments, vitals deltas, resource view). 3. Harden **event validation** and add integration tests. 4. Run a **shadow study** with clinicians: measure acceptance/fatigue and revise thresholds. 5. Prepare de-identified **replay dataset** to replace synthetic events.

Bottom line: We turned a fragile monolith toward a **skill-based, governance-first workflow assistant**, re-anchored in your domain expertise, with audits to stop future regression.